

Lab 1. Environment Setup & Java Basics

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2.2.1 Write, compile the first Java application:

Code:

```
1 public class Ex_2_2_1 {  
    Run | Debug  
2     public static void main(String args[]) {  
3         System.out.println(x: "Xin chao \n cac ban!");  
4         System.out.println(x: "Hello \t world!");  
5     }  
6 }
```

Result:

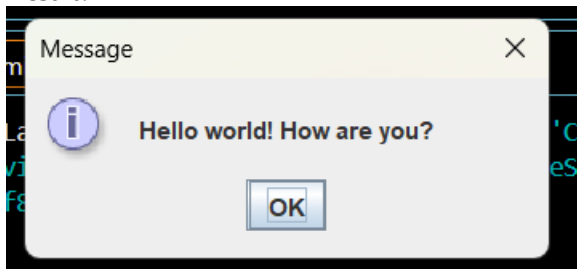
```
Xin chao  
cac ban!  
Hello    world!
```

2.2.2 Write, compile the first dialog Java program

Code:

```
1 import javax.swing.JOptionPane;  
2  
3 public class Ex_2_2_2 {  
    Run | Debug  
4     public static void main(String[] args) {  
5         JOptionPane.showMessageDialog(parentComponent: null, message: "Hello world! How are you?");  
6         System.exit(status: 0);  
7     }  
8 }
```

Result:

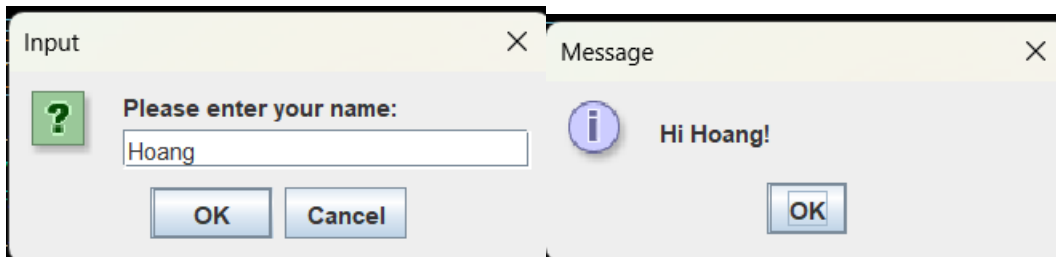


2.2.3 Write, compile the first input dialog Java application

Code:

```
1 import javax.swing.JOptionPane;
2
3 public class Ex_2_2_3 {
4     Run | Debug
5     public static void main(String[] args) {
6         String result;
7         result = JOptionPane.showInputDialog(message: "Please enter your name:");
8         JOptionPane.showMessageDialog(parentComponent: null, "Hi " + result + "!");
9         System.exit(status: 0);
10    }
```

Result:

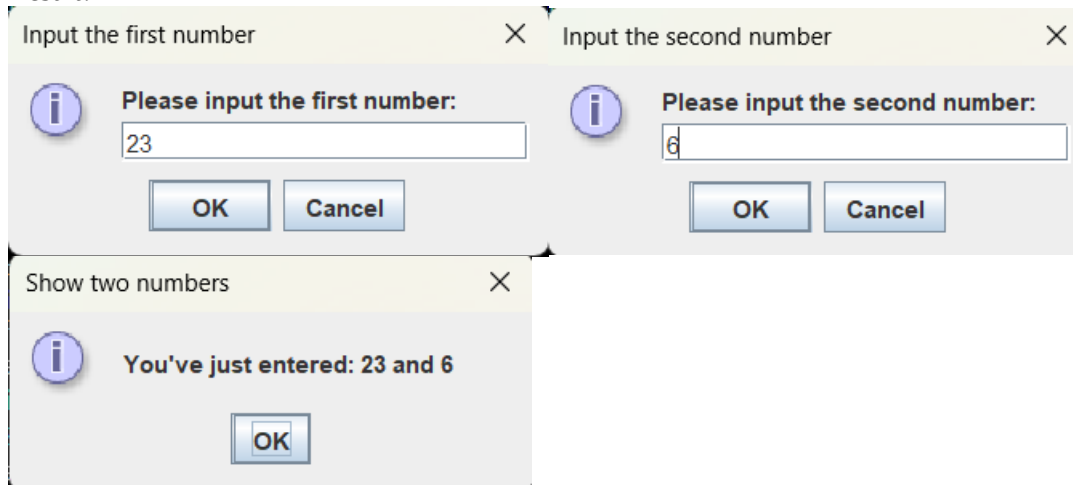


2.2.4 Write, compile, and run the following example:

Code:

```
1 import javax.swing.JOptionPane;
2
3 public class Ex_2_2_4 {
4     Run | Debug
5     public static void main(String[] args) {
6         String strNum1, strNum2;
7         String strNotification = "You've just entered: ";
8
9         strNum1 = JOptionPane.showInputDialog(parentComponent: null,
10         |     message: "Please input the first number: ", title: "Input the first number",
11         |     JOptionPane.INFORMATION_MESSAGE);
12         strNotification += strNum1 + " and ";
13
14         strNum2 = JOptionPane.showInputDialog(parentComponent: null,
15         |     message: "Please input the second number: ", title: "Input the second number",
16         |     JOptionPane.INFORMATION_MESSAGE);
17         strNotification += strNum2;
18
19         JOptionPane.showMessageDialog(parentComponent: null, strNotification,
20         |     title: "Show two numbers", JOptionPane.INFORMATION_MESSAGE);
21         System.exit(status: 0);
22     }
23 }
```

Result:



2.2.5 Write a program to calculate sum, difference, product, and quotient of 2 double numbers which are entered by users.

Notes

- To convert from String to double, you can use

`double num1 = Double.parseDouble(strNum1)`

- Check the divisor of the division

Code:

```
1  import java.util.*;
2
3  public class Ex_2_2_5 {
4      Run | Debug
5      public static void main(String[] args) {
6          Scanner scanner = new Scanner(System.in);
7
8          System.out.print(s: "Enter first number: ");
9          double a = scanner.nextDouble();
10         System.out.print(s: "Enter second number: ");
11         double b = scanner.nextDouble();
12
13         double sum = a + b;
14         System.out.println("The sum is " + sum);
15
16         double diff = Math.abs(a - b);
17         System.out.println("The difference is " + diff);
18
19         double prod = a * b;
20         System.out.println("The product is " + prod);
21
22         String x = "The sum is " + sum + "\nThe difference is " + diff +
23         |           "\nThe product is " + prod;
24         if (b != 0) {
25             double quot = a / b;
26             x += "\nThe quotient is " + quot;
27         } else {
28             x += "\nCannot divide by zero!!!";
29         }
30         System.out.println(x);
31         scanner.close();
32     }
33 }
34
```

Result:

```
Enter first number: 234      Enter first number: 98
Enter second number: -23    Enter second number: 0
The sum is 211.0            The sum is 98.0
The difference is 257.0     The difference is 98.0
The product is -5382.0      The product is 0.0
The sum is 211.0            The sum is 98.0
The difference is 257.0     The difference is 98.0
The product is -5382.0      The product is 0.0
The quotient is -10.173913043478262  Cannot divide by zero!!!
```

2.2.6 Write a program to solve:

For simplicity, we only consider the real roots of the equations in this task.

- The first-degree equation (linear equation) with one variable

Note: A first-degree equation with one variable can have a form such as $ax + b = 0$ ($a \neq 0$).

You should handle the case where the user input value 0 for a.

- The system of first-degree equations (linear system) with two variables

Note: A system of first-degree equations with two variables x_1 and x_2 can be written as follows.

$$\begin{cases} a_{11}x_1 + a_{12}x_2 = b_1 \\ a_{21}x_1 + a_{22}x_2 = b_2 \end{cases}$$

You should handle the case where the values of the coefficients produce infinitely many solutions and the case where they produce no solution.

Hint:

Use the following determinants:

$$\begin{aligned} D &= \begin{vmatrix} a_{11} & a_{12} & a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{21}a_{12} \\ D_1 &= \begin{vmatrix} b_1 & a_{12} & b_2 & a_{22} \end{vmatrix} = b_1a_{22} - b_2a_{12} \\ D_2 &= \begin{vmatrix} a_{11} & b_1 & a_{21} & b_2 \end{vmatrix} \\ &= a_{11}b_2 - a_{21}b_1 \end{aligned}$$

- The second-degree equation with one variable

Note: A second-degree equation with one variable (i.e., quadratic equation) can have a form such as $ax^2 + bx + c = 0$, where x is the variable, and a , b , and c are coefficients ($a \neq 0$).

You should handle the case where the values of the coefficients produce a double root & the case where they produce no root. You should also handle the case where the user input value 0 for a .

Hint:

Use the discriminant $\Delta = b^2 - 4ac$

Code:

```
1  import java.util.*;
2
3  public class Ex_2_2_6 {
4      Run | Debug
5      public static void main(String[] args) {
6
7          Scanner sc = new Scanner(System.in);
8
9          int deg = 0;
10
11         while (true) {
12             System.out.print("Enter the degree of equation (1 or 2): ");
13             int input1 = sc.nextInt();
14             if (input1 == 1 || input1 == 2) {
15                 deg = input1;
16                 break;
17             } else {
18                 System.out.println("Invalid input, try again.");
19             }
20         }
21
22         int variable = 0;
```

```
22
23 while (true) {
24     if (deg == 1) {
25         System.out.print("Enter the number of variables (1 or 2): ");
26         int input2 = sc.nextInt();
27         if (input2 == 1 || input2 == 2) {
28             variable = input2;
29             break;
30         } else {
31             System.out.println("Invalid input, try again.");
32         }
33     } else { x:
34         variable = 1;
35         break;
36     }
37 }
38
39 System.out.println("Degree selected: " + deg);
40 System.out.println("Number of variables: " + variable);
41
```

```
42 // Case 1: Linear, 1 variable
43 if (deg == 1 && variable == 1) {
44     int a;
45     while (true) {
46         System.out.print("Enter a (for  $ax + b = 0$ ): ");
47         int a0 = sc.nextInt();
48         if (a0 != 0) {s:
49             a = a0;
50             break;
51         } else {
52             System.out.println("Invalid input, a must differ from 0, try again.");
53         }
54     }x:
55     System.out.print("Enter b (for  $ax + b = 0$ ): ");
56     int b = sc.nextInt();
57     double result = -s:(double)b / a;
58     System.out.println("The solution is x = " + result);
59 }
```

```
61 // Case 2: Linear, 2 variables
62 if (deg == 1 && variable == 2) {
63     System.out.print("Enter a11 (for a11*x1 + a12*x2 = b1): ");
64     int a11 = sc.nextInt();
65     System.out.print(s: "Enter a12 (for a11*x1 + a12*x2 = b1): ");
66     int a12 = sc.nextInt();
67     System.out.print(s: "Enter b1 (for a11*x1 + a12*x2 = b1): ");
68     int b1 = sc.nextInt();
69     System.out.print(s: "Enter a21 (for a21*x1 + a22*x2 = b2): ");
70     int a21 = sc.nextInt();
71     System.out.print(s: "Enter a22 (for a21*x1 + a22*x2 = b2): ");
72     int a22 = sc.nextInt();
73     System.out.print(s: "Enter b2 (for a21*x1 + a22*x2 = b2): ");
74     int b2 = sc.nextInt();
75     s:
76     int D = a11 * a22 - a12 * a21;
77     int D1 = b1 * a22 - b2 * a12;
78     int D2 = a11 * b2 - a21 * b1;
79
80     if (D != 0) {
81         double x1 = (double) D1 / D;
82         double x2 = (double) D2 / D;
83         System.out.println("Unique solution:");
84         System.out.println("x1 = " + x1);
85         System.out.println(x: "x2 = " + x2);
86     } else {
87         if (D1 == 0 && D2 == 0) {
88             System.out.println("The system has infinitely many solutions.");
89         } else {
90             System.out.println(x: "The system has no solution.");
91         }
92     } x:
93 }
94
95 // Case 3: Quadratic, 1 variable
96 if (deg == 2 && variable == 1) {
97     int a;
98     while (true) {
99         System.out.print("Enter a (for a*x^2 + b*x + c = 0): ");
100         int a0 = sc.nextInt();
101         if (a0 != 0) {s:
102             a = a0;
103             break;
104         } else {
105             System.out.println("Invalid input, a must differ from 0, try again.");
106         }
107     } x:
108
109     System.out.print("Enter b (for a*x^2 + b*x + c = 0): ");
110     int b = sc.nextInt();
111     s:
112     System.out.print("Enter c (for a*x^2 + b*x + c = 0): ");
113     int c = sc.nextInt();
114     s:
115     double delta = b * b - 4.0 * a * c;
```

```
116
117     if (delta == 0) {
118         double x = - (double)b / (2 * a);
119         System.out.println("The equation has one double root:");
120         System.out.println("x = " + x);
121     } else if (delta > 0) {x:
122         double x1 = (-b + Math.sqrt(delta)) / (2 * a);
123         double x2 = (-b - Math.sqrt(delta)) / (2 * a);
124         System.out.println("The equation has two distinct real roots:");
125         System.out.println("x1 = " + x1);
126         System.out.println(x: "x2 = " + x2);
127     } else {
128         System.out.println("The equation has no real solution.");
129     }
130 }x:
131
132     sc.close();
133 }
134 }
```

Result:

First case: (when user enters the value of a is 0, the notification appears and returns to enter input again)

```
Enter the degree of equation (1 or 2): 1
Enter the number of variables (1 or 2): 1
Degree selected: 1
Number of variables: 1
Enter a (for  $ax + b = 0$ ): 0
Invalid input, a must differ from 0, try again.
Enter a (for  $ax + b = 0$ ): 1
Enter b (for  $ax + b = 0$ ): -3
The solution is  $x = 3.0$ 
```

Second case:

```
Enter the degree of equation (1 or 2): 1
Enter the number of variables (1 or 2): 2
Degree selected: 1
Number of variables: 2
Enter a11 (for  $a_{11}x_1 + a_{12}x_2 = b_1$ ): 1
Enter a12 (for  $a_{11}x_1 + a_{12}x_2 = b_1$ ): 2
Enter b1 (for  $a_{11}x_1 + a_{12}x_2 = b_1$ ): 1
Enter a21 (for  $a_{21}x_1 + a_{22}x_2 = b_2$ ): 2
Enter a22 (for  $a_{21}x_1 + a_{22}x_2 = b_2$ ): 3
Enter b2 (for  $a_{21}x_1 + a_{22}x_2 = b_2$ ): 4
Unique solution:
x1 = 5.0
x2 = -2.0
```

```
Enter the degree of equation (1 or 2): 1
Enter the number of variables (1 or 2): 2
Degree selected: 1
Number of variables: 2
Enter a11 (for  $a_{11}x_1 + a_{12}x_2 = b_1$ ): 1
Enter a12 (for  $a_{11}x_1 + a_{12}x_2 = b_1$ ): 1
Enter b1 (for  $a_{11}x_1 + a_{12}x_2 = b_1$ ): 2
Enter a21 (for  $a_{21}x_1 + a_{22}x_2 = b_2$ ): 1
Enter a22 (for  $a_{21}x_1 + a_{22}x_2 = b_2$ ): 1
Enter b2 (for  $a_{21}x_1 + a_{22}x_2 = b_2$ ): 3
The system has no solution.

Enter the degree of equation (1 or 2): 1
Enter the number of variables (1 or 2): 2
Degree selected: 1
Number of variables: 2
Enter a11 (for  $a_{11}x_1 + a_{12}x_2 = b_1$ ): 1
Enter a12 (for  $a_{11}x_1 + a_{12}x_2 = b_1$ ): 2
Enter b1 (for  $a_{11}x_1 + a_{12}x_2 = b_1$ ): 1
Enter a21 (for  $a_{21}x_1 + a_{22}x_2 = b_2$ ): 2
Enter a22 (for  $a_{21}x_1 + a_{22}x_2 = b_2$ ): 4
Enter b2 (for  $a_{21}x_1 + a_{22}x_2 = b_2$ ): 2
```

Third case:

```
Enter the degree of equation (1 or 2): 2
Degree selected: 2
Number of variables: 1
Enter a (for  $a x^2 + b x + c = 0$ ): 1
Enter b (for  $a x^2 + b x + c = 0$ ): 1
Enter c (for  $a x^2 + b x + c = 0$ ): 1
The equation has no real solution.

Enter the degree of equation (1 or 2): 2
Degree selected: 2
Number of variables: 1
Enter a (for  $a x^2 + b x + c = 0$ ): 1
Enter b (for  $a x^2 + b x + c = 0$ ): -4
Enter c (for  $a x^2 + b x + c = 0$ ): 4
The equation has one double root:
x = 2.0
```

(when user enters the value of a is 0, the notification appears and returns to enter input again)

```
Enter the degree of equation (1 or 2): 2
Degree selected: 2
Number of variables: 1
Enter a (for  $a*x^2 + b*x + c = 0$ ): 0
Invalid input, a must differ from 0, try again.
Enter a (for  $a*x^2 + b*x + c = 0$ ): 1
Enter b (for  $a*x^2 + b*x + c = 0$ ): -5
Enter c (for  $a*x^2 + b*x + c = 0$ ): 6
The equation has two distinct real roots:
x1 = 3.0
x2 = 2.0
```

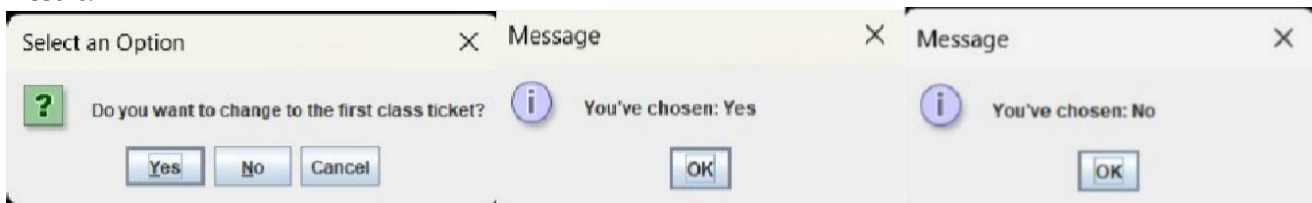
6.1 Write, compile and run the ChoosingOption program:

Note: We use JavaBasics project for this exercise.

Code:

```
1  import javax.swing.JOptionPane;
2
3  public class Ex_6_1_ChoosingOption {
4      Run | Debug
4      public static void main(String[] args) {
5          int option = JOptionPane.showConfirmDialog(parentComponent: null,
6              message: "Do you want to change to the first class ticket?");
7
8          JOptionPane.showMessageDialog(parentComponent: null, "You've chosen: "
9              + (option==JOptionPane.YES_OPTION?"Yes":"No"));
10         System.exit(status: 0);
11     }
12 }
```

Result:



6.2 Write a program for input/output from keyboard

Note: We use the JavaBasics project for this exercise.

Code:

```
import java.util.Scanner;

public class Ex_6_2_InputFromKeyboard {
    Run | Debug
    public static void main(String[] args) {
        Scanner keyboard = new Scanner(System.in);

        System.out.println(x: "What's your name?");
        String strName = keyboard.nextLine();
        System.out.println(x: "How old are you?");
        int iAge = keyboard.nextInt();
        System.out.println(x: "How tall are you (m)?");
        double dHeight = keyboard.nextDouble();

        System.out.println("Mrs/Ms. " + strName + ", " + iAge + " years old. "
            + "Your height is " + dHeight + ".");
        keyboard.close();
    }
}
```

Result:

```
What's your name?
Hoang
How old are you?
20
How tall are you (m)?
1.72
Mrs/Ms. Hoang, 20 years old. Your height is 1.72.
```

Question: What happen if users choose "Cancel"?

*In the program, JOptionPane.showConfirmDialog() will return one of three cases:

-JOptionPane.YES_OPTION (0) if choosing YES

-JOptionPane.NO_OPTION (1) if choosing NO

JOptionPane.CANCEL_OPTION (2) if choosing CANCEL

*But the program will check JOptionPane.YES_OPTION and recognize that other choices is NO. So, if users choose CANCEL,

the program shows the result: "You've chosen: No"

Question: How to customize the options to users, e.g only two options: "Yes" and "No", OR " I do" and "I don't" (Suggestion: Use JavaDocs or using Eclipse/Netbean IDE help)

*By default, JOptionPane.showConfirmDialog() displays 3 options: Yes, No, Cancel. To limit to only two options: "Yes", "No"

we can use the following:

-Only show "Yes"; "No"

import

javax.swing.JOptionPane;

```
public class YesNoDialog {
    public static void main(String[] args) {
        int choice = JOptionPane.showOptionDialog(
            null,
            "Do you like
            it?", "Your
            feedback",
            JOptionPane.YES_NO_OPTION,
            JOptionPane.QUESTION_MESSAGE
            , null,
            null
            ,
            null
        );

        if (choice == 0) {
            JOptionPane.showMessageDialog(null, "User selected: Yes");
        } else if (choice == 1) {
            JOptionPane.showMessageDialog(null, "User selected: No");
        } else {
            JOptionPane.showMessageDialog(null, "User closed the dialog");
        }
    }
}
```

-To customize the button names instead of "Yes"; "No". We pass an Object[] array containing the button name to showOptionDialog()

```
import  
  
javax.swing.JOptionPane;  
  
public class CustomizeDialog  
{  
    public static void main(String[] args) {  
        Object[] options = {"I do", "I don't"};  
        int choice = JOptionPane.showOptionDialog(  
            null,
```

```
"Do you like  
it?", "Your feed  
back",  
JOptionPane.YES_NO_OPTION,  
JOptionPane.QUESTION_MESSAGE  
, null,  
options,  
options[0]  
);  
  
if (choice == 0) {  
    JOptionPane.showMessageDialog(null, "User selected: I do");  
} else if (choice == 1) {  
    JOptionPane.showMessageDialog(null, "User selected: I don't");  
} else {  
    JOptionPane.showMessageDialog(null, "User closed the dialog");  
}  
}  
}
```

6.3 Write a program to display a triangle with a height of n stars (*), n is entered by users.

E.g. $n=5$:

```
  *
 * * *
* * * * *
* * * * *
* * * * *
```

Code:

```
1  import java.util.*;
2
3  public class Ex_6_3 {
4      Run | Debug
5      public static void main(String[] args) {
6          Scanner scanner = new Scanner(System.in);
7          System.out.print(s: "Enter the height of the triangle: ");
8          int n = scanner.nextInt();
9
10         for (int i = 1; i <= n; i++) {
11             for (int j = 1; j <= n - i; j++) {
12                 System.out.print(s: " ");
13             }
14             for (int j = 1; j <= 2 * i - 1; j++) {
15                 System.out.print(s: "*");
16             }
17             System.out.println();
18         }
19         scanner.close();
20     }
21 }
```

Result:

```
Enter the height of the triangle: 6
 *
 * * *
 * * * * *
 * * * * *
 * * * * *
 * * * * *
```

6.4 Write a program to display the number of days of a month, which is entered by users (both month and year). If it is an invalid month/year, ask the user to enter again.

Note: You must create a new Java project for this exercise.

- The user can either enter a month in its full name, abbreviation, in 3 letters, or in number. To illustrate, the valid inputs of *January* are January, Jan., Jan, and 1.
- The user must enter a year in a non-negative number and enter all the digits. For instance, the valid inputs of year *1999* is only 1999, but not 99, "one thousand nine hundred ninety-nine", or anything else.
- A year is either a common year of 365 days or a leap year of 366 days. Every year that is divisible by 4 is a leap year, except for years that are divisible by 100, but not by 400. For instance, year 1800 is not a leap year, yet year 2000 is a leap year. In a year, there are twelve months, which are listed in order as follows.

Month	January	February	March	April	May	June	July	August	September	October	November	December
Abbreviation	Jan.	Feb.	Mar.	Apr.	May	June	July	Aug.	Sept.	Oct.	Nov.	Dec.
In 3 letters	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
In Number	1	2	3	4	5	6	7	8	9	10	11	12
Days of Month in Common Year	31	28	31	30	31	30	31	31	30	31	30	31
Days of Month in Leap Year	31	29	31	30	31	30	31	31	30	31	30	31

Code:

```

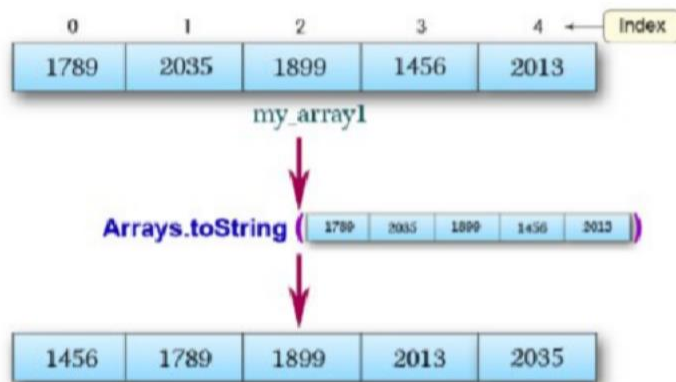
1  import java.util.*;
2
3  public class Ex_6_4 {
4      public static void main(String[] args) {
5          Run | Debug
6          Scanner sc = new Scanner(System.in);
7
8          String[][] months = {
9              {"1", "jan", "jan.", "january"},
10             {"2", "feb", "feb.", "february"},
11             {"3", "mar", "mar.", "march"},
12             {"4", "apr", "apr.", "april"},
13             {"5", "may", "may."},
14             {"6", "jun", "jun.", "june"},
15             {"7", "jul", "jul.", "july"},
16             {"8", "aug", "aug.", "august"},
17             {"9", "sep", "sep.", "september"},
18             {"10", "oct", "oct.", "october"},
19             {"11", "nov", "nov.", "november"},
20             {"12", "dec", "dec.", "december"}
21         };
22     }
23 }
```

```
22     int month = -1, year = -1;
23
24     while (month == -1) {
25         System.out.print("Enter month: ");
26         String input = scs.nextLine().toLowerCase().trim();
27         for (String[] m : months)
28             for (String alias : m)
29                 if (alias.equals(input))
30                     month = Integer.parseInt(m[0]);
31         if (month == -1)
32             System.out.println("Invalid month, try again.");
33     }x:
34
35     while (year < 0) {
36         System.out.print("Enter year: ");
37         String input = scs.nextLine().trim();
38         if (input.matches("\\d+"))
39             year = Integer.parseInt(input);
40         else
41             System.out.println("Invalid year, try again.");
42     }x:
44     int[] days = {31,28,31,30,31,30,31,31,30,31,30,31};
45     if (month == 2 && ((year%4==0 && year%100!=0) || year%400==0))
46         days[1] = 29;
47
48     System.out.println("Month " + month + " of year " + year + " has " + days[month-1] + " days.");
49     sc.close();
50 }
51 }
```

Result:

Enter month: feb	Enter month: 2
Enter year: 2024	Enter year: 2025
Month 2 of year 2024 has 29 days.	Month 2 of year 2025 has 28 days.

6.5 Write a Java program to sort a numeric array, and calculate the sum and average value of array elements.



Note: You must create a new Java project for this exercise.

- The array can be entered by the user or a constant.

Code:

```
1  import java.util.Scanner;
2
3  public class Ex_6_5 {
4      Run | Debug
5      public static void main(String[] args) {
6          Scanner sc = new Scanner(System.in);
7          try {
8              System.out.print(s: "Enter the number of elements: ");
9              int n = sc.nextInt();
10             int[] numbers = new int[n];
11             System.out.print("Enter '" + n + "' integers: ");
12             for (int i = 0; i < n; i++) {
13                 numbers[i] = sc.nextInt();
14             }
15             sortArray(numbers);
16             System.out.print(s: "Sorted array: ");
17             for (int num : numbers) {
18                 System.out.print(num + " ");
19             }
20             System.out.println();
21             System.out.println("Sum: " + sumArray(numbers));
22             System.out.println("Average: " + averArray(numbers));
23         } finally {
24             sc.close();
25         }
26     }
27 }
```

```
27 public static void sortArray(int[] array) {
28     for (int i = 0; i < array.length - 1; i++) {
29         for (int j = i + 1; j < array.length; j++) {
30             if (array[i] > array[j]) {
31                 int temp = array[i];
32                 array[i] = array[j];
33                 array[j] = temp;
34             }
35         }
36     }
37 }
38
39 public static int sumArray(int[] array) {
40     int sum = 0;
41     for (int i = 0; i < array.length; i++) {
42         sum = sum + array[i];
43     }
44     return sum;
45 }
46
47 public static int averArray(int[] array) {
48     int avg = sumArray(array) / array.length;
49     return avg;
50 }
51 }
```

Result:

```
Enter the number of elements: 5
Enter '5' integers: 34 36 45 6 -1
Sorted array: -1 6 34 36 45
Sum: 120
Average: 24
```

6.6 Write a Java program to add two matrices of the same size.

Note: You must create a new Java project for this exercise.

- The matrices can be entered by the user or constants.

Code:

```
1  import java.util.Scanner;
2
3  public class Ex_6_6 {
4      Run | Debug
5      public static void main(String[] args) {
6          Scanner scanner = new Scanner(System.in);
7
8          System.out.print(s: "Enter the column size of matrix: ");
9          int col = scanner.nextInt();
10         System.out.print(s: "Enter the row size of matrix: ");
11         int row= scanner.nextInt();
12
13         int[][] matrix1 = new int[row][col];
14         int[][] matrix2 = new int[row][col];
15         int[][] result = new int[row][col];
16
17         System.out.println(x: "Enter elements of first matrix:");
18         for (int i = 0; i < row; i++) {
19             for (int j = 0; j < col; j++) {
20                 matrix1[i][j] = scanner.nextInt();
21             }
22         }
```

```
24      System.out.println(x: "Enter elements of second matrix:");
25      for (int i = 0; i < row; i++) {
26          for (int j = 0; j < col; j++) {
27              matrix2[i][j] = scanner.nextInt();
28          }
29      }
30
31      for (int i = 0; i < row; i++) {
32          for (int j = 0; j < col; j++) {
33              result[i][j] = matrix1[i][j] + matrix2[i][j];
34          }
35      }
36
37      System.out.println("The result is: ");
38      for (int i = 0; i < row; i++) {
39          for (int j = 0; j < col; j++) {
40              System.out.print(result[i][j] + " ");
41          }
42          System.out.println();
43      }
44
45      scanner.close();
46  }
47 }
```

Result:

```
Enter the column size of matrix: 2
Enter the row size of matrix: 3
Enter elements of first matrix:
2 3
5 6
-1 2
Enter elements of second matrix:
3 4
7 8
9 11
The result is:
5 7
12 14
8 13
```

